Matthew Doyle

Toronto, Ontario | matthew.doyle@icloud.com | website | linkedin

Projects

Order Flow Imbalance Signal **2026–Present**

Python, pandas, scikit-learn, XGBoost, SQL, DuckDB, DuckLake, IBKR, Git

- Developed a medium-horizon, dollar-neutral long-short strategy for S&P 500 equities using proprietary order flow data from ExTech.
- Built a version-controlled pipeline for data ingestion, feature engineering, model training, portfolio construction, backtesting and paper execution.
- Produced a three-year out-of-sample backtest with 13.9% ANR, 1.61 Sharpe and 6.1% maximum drawdown after estimated transaction costs.
- Built a PCA risk model and hedge portfolio for systematic risk management, producing approximately zero market beta.

Binary Option Arbitrage **2026–Present**

Python, Redis, data engineering, Databento, trading APIs, derivatives, trade execution

- Built a research and execution system for binary-option arbitrage across prediction and traditional financial markets.
- Collected and normalized market and order-book data to identify theoretically risk-free opportunities using a multi-leg derivatives strategy.
- Automated simultaneous order execution, fill monitoring, position reconciliation and execution-risk management through trading APIs.
- Deployed the strategy with live capital, generating an annualized money-weighted return of approximately 40% since inception.

Professional Experience

The Entrepreneurship Hatchery, University of Toronto **Toronto, Ontario**

Venture Associate (contract) *Sep 2025–Mar 2026*

- Sourced dealflow for the accelerator by screening technology projects and founders in the University of Toronto engineering research community.
- Conducted commercial diligence on early-stage technology ventures, assessing market opportunity, positioning, and team/founder fit.
- Collaborated with researchers, founders, mentors and investors to deliver programs across the University of Toronto ecosystem.

Higher Education Strategy Associates **Toronto, Ontario**

Research Analyst *May 2024–Sep 2025*

- Consulted on academic-program viability through analysis of enrolment, labour-market demand, industry hiring patterns and program curricula.
- Analyzed provincial funding programs and institutional financial data to support client recommendations and strategic planning.
- Built a data pipeline that amalgamated news feeds using AI and NLP, to generate structured data and insights into financial state of the sector.

Students Nova Scotia **Halifax, Nova Scotia**

Chair, Board of Directors *May 2023–Apr 2024*

- Oversaw board governance and organizational strategy, liaising with member organizations, government officials and external stakeholders.
- Helped secure over $16 million in financial-aid grants through policy research, government advocacy and stakeholder engagement.

St. Francis Xavier Students' Union **Antigonish, Nova Scotia**

Vice President, External Affairs *May 2023–Apr 2024*

- Led research and advocacy on student housing, financial aid and education policy, developing recommendations for government stakeholders.
- Advanced reforms enabling post-secondary institutions to access CMHC financing for student housing and institutional development.
- Represented students before provincial and national policymakers while coordinating government relations and sector partnerships.

Education & Credentials

St. Francis Xavier University **Antigonish, Nova Scotia**

Bachelor of Arts, First Class Honours with Distinction, Philosophy *Sep 2019–May 2024*

- Wrote an honours thesis on the normative and applied ethics of the alignment problem in artificial intelligence (AI).
- **Awards and Achievements:** Dean's List (2023, 2024), Father Charles R. MacDonald Memorial Medal in Philosophy (2024), Mulroney Institute of Public Policy and Governance Case Competition (1st place 2024), Students Nova Scotia Most Valuable Player Award (2024), Canadian Alliance of Student Associations Advocacy Expert Award (2024), Golden X Leadership Award (2024), LSAT Score: 176

Credentials

CFA Program — Level I Candidate, Nov 2026

Python for Data Science — University of Michigan

Options 101 — Akuna Capital

Skills

Technical: Python, SQL, Excel, Git, version control, pandas, NumPy, scikit-learn, XGBoost, DuckDB, Docker, Google Cloud, OpenAI API, Git, LaTeX.

Quantitative research: Applied mathematics, backtesting, data pipeline, order-flow analysis, derivatives, risk management, portfolio management, signal research, trade execution and model evaluation.

General research: Capital markets, financial markets, financial analysis, investment memos, policy research, problem-solving and expert interviews.